

Todo/Task

```
> █ Claude Code CLI ███ Todo/Task ██████████
```

> ■■■■: 2026-07-07

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1. ██████████
2. ██████████ Todo V1 █ Task V2
3. TodoWriteTool████V1████
4. Task ██████V2████
5. ██████Sub-agent████ Task ██████████
6. ██████████
7. ██████████Background Tasks█
8. ████████ UI ███
9. ██████████
10. ██████████████
11. ██████████

1. ■■■■■■

Todo/Task

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- [illegible]

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■■■■■ ①-③ ■■■■■■■■■■ ③ ■■■■ agent ■■ todo ■■■■■■■■■■■■■■■■■■■■

```

TaskUpdate
taskId: string → ID

```

5/1/2014

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- ### 4.3 TaskListTool — ■■■■

— 100 —

- ## 4.4 TaskGetTool —

4.5 ■■■■ Prompt ■■■■

4.5.1 TaskCreateTool ■■■■

Prompt■■■■■■■■ LLM ■■■■■■:

■■■■■■■■■■

- ## 1. ■■■■When to Use This Tool■■■

- `taskStatus` — 3 `enum`
- `taskStatus` — `enum`
- Plan mode `enum`
- `taskStatus` todo list
- `taskStatus`
- `taskStatus` TaskCreate `enum`
- `taskStatus` in_progress
- `taskStatus`

2. `taskStatus` When NOT to Use `enum`

- `taskStatus` 1 `enum`
- `taskStatus`
- `taskStatus` 3 `enum`
- `taskStatus` / `enum`

3. `taskStatus`

- subject `enum`
- description `enum`
- activeForm `enum` spinner `enum`

4. `taskStatus`

- `taskStatus`
- `taskStatus` TaskUpdate `enum`
- `taskStatus` TaskList `enum`

5. `taskStatus` isAgentSwarmsEnabled() `enum`

- `taskStatus` "description `enum` agent `enum`"
- `taskStatus` "pending `enum` owner `enum` TaskUpdate `enum` owner `enum`"

4.5.2 TaskUpdateTool `enum`

`enum`: src/tools/TaskUpdateTool/prompt.ts

Description: Update a task in the task list

Prompt `enum`

1. `taskStatus`
2. `taskStatus` deleted `enum`
3. `taskStatus` status, subject, description, activeForm, owner, metadata, addBlocks, addBlockedBy
4. `taskStatus` pending → in_progress → completed `enum` deleted `enum`
5. `taskStatus` TaskGet `enum`
6. JSON `enum` 5 `enum` JSON `enum`

4.5.3 TaskGetTool `enum`

`enum`: src/tools/TaskGetTool/prompt.ts

Description: Get a task by ID from the task list

Prompt `enum`

- `taskStatus`
- `taskStatus` blocks/blockedBy `enum`
- `taskStatus`
- `taskStatus` subject, description, status, blocks, blockedBy

- `taskBlockedBy` blockedBy `taskBlockedBy`

4.5.4 TaskListTool `TaskListTool`

`TaskListTool`: `src/tools/TaskListTool/prompt.ts`

Description: List all tasks in the task list

Prompt `TaskListTool`

- `taskBlockedBy` status=pending, `taskBlockedBy` owner, `taskBlockedBy`
- `taskBlockedBy`
- `taskBlockedBy`
- `taskBlockedBy` ID `taskBlockedBy`
- `taskBlockedBy` id, subject, status, owner, blockedBy

`TaskListTool` isAgentSwarmsEnabled() `TaskListTool`

1. `TaskList` `TaskList`

2. `status=pending`, `owner`, empty `blockedBy` `TaskList`

3. `ID` `TaskList`

4. `TaskUpdate` `TaskList` (`owner`)

5. `TaskList`

4.6 `TaskListTool` Sub-agent `TaskListTool` Task `TaskListTool`

> `TaskListTool` 5 `TaskListTool`

`TaskListTool`

- `TaskListTool` — `TaskListTool` Task `TaskListTool`
- `TaskListTool` — `TaskListTool` Task `TaskListTool`
- In-process teammate — `TaskListTool` Task `TaskListTool`
- `TaskListTool` task list `TaskListTool` session ID `TaskListTool`

4.7 TaskCreate `TaskCreate` vs TodoWrite `TaskCreate`

> `TaskCreate` 5.5 `TaskCreate`

<code>TaskCreate</code>	V1 TodoWriteTool	V2 TaskCreateTool
<code>TaskCreate</code>	<code>TaskCreate</code>	<code>TaskCreate</code>
<code>TaskCreate</code>	<code>TaskCreate</code>	<code>TaskUpdate</code>
<code>TaskCreate</code>	1 <code>TaskCreate</code> = <code>TaskCreate</code>	8 <code>TaskCreate</code> = 8 <code>TaskCreate</code>

5. `TaskCreate` Sub-agent `TaskCreate` Task `TaskCreate`

5.1 `TaskCreate`

`TaskCreate` Task `TaskCreate`

<code>TaskCreate</code>	<code>TaskCreate/Update/Get/List?</code>	<code>TaskCreate</code>
<code>**TaskCreate**</code> <code>Explore</code> <code>Plan</code> <code>code-reviewer</code>	<code>TaskCreate</code>	<code>ALL_AGENT_DISALLOWED_TOOLS</code>
<code>**TaskCreate**</code> <code>run_in_background</code> <code>=true</code>	<code>TaskCreate</code>	<code>ASYNC_AGENT_ALLOWED_TOOLS</code>
<code>**In-process teammate**</code> <code>TaskCreate</code>	<code>TaskCreate</code>	<code>IN_PROCESS_TEAMMATE_ALLOWED_TOOLS</code>

■■■■■	■■■ TaskCreate/Update/Get/List?	■■
■■■■■■■agent	■	■■■■■■■■■

5.2 ■■■■■■

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5.3 ■■■

```
■■: src/constants/tools.ts
```

```
// █ agent ██████████Task ████████ → ██████
// AGENT_TOOL_NAME █ ant ██████████WORKFLOW_TOOL_NAME █ feature flag ██████
export const ALL_AGENT_DISALLOWED_TOOLS = new Set([
  TASK_OUTPUT_TOOL_NAME, // █
  EXIT_PLAN_MODE_V2_TOOL_NAME, // █
  ENTER_PLAN_MODE_TOOL_NAME, // █
  ...(process.env.USER_TYPE === 'ant' ? [] : [AGENT_TOOL_NAME]),
  ASK_USER_QUESTION_TOOL_NAME, // █
  TASK_STOP_TOOL_NAME, // █
  ...(feature('WORKFLOW_SCRIPTS') ? [WORKFLOW_TOOL_NAME] : []),
])

// █ agent ██████Task ████████ → █
export const ASYNC_AGENT_ALLOWED_TOOLS = new Set([
  FILE_READ_TOOL_NAME, WEB_SEARCH_TOOL_NAME, TODO_WRITE_TOOL_NAME, ...
])

// In-process teammate ██████████
export const IN_PROCESS_TEAMMATE_ALLOWED_TOOLS = new Set([
  TASK_CREATE_TOOL_NAME, TASK_GET_TOOL_NAME,
  TASK_LIST_TOOL_NAME, TASK_UPDATE_TOOL_NAME,
  SEND_MESSAGE_TOOL_NAME,
])
```

```

■■: src/tools/AgentTool/agentToolUtils.ts■■■■■■■■■■

```

```
export function filterToolsForAgent({ tools, isBuiltIn, isAsync, permissionMode }) {
  return tools.filter(tool => {
    // MCP agent
    if (tool.name.startsWith('mcp_')) return true
    // Plan ExitPlanMode
    if (toolMatchesName(tool, EXIT_PLAN_MODE_V2_TOOL_NAME) && permissionMode === 'plan') return true
    if (ALL_AGENT_DISALLOWED_TOOLS.has(tool.name)) return false
  })
}
```



```
if (!isBuiltIn && CUSTOM_AGENT_DISALLOWED_TOOLS.has(tool.name)) return false
if (isAsync && !ASYNC_AGENT_ALLOWED_TOOLS.has(tool.name)) {
  if (isAgentSwarmsEnabled() && isInProcessTeammate()) {
    if (IN_PROCESS_TEAMMATE_ALLOWED_TOOLS.has(tool.name)) return true
  }
  return false // agent
}
return true
})
}
```

5.4 Task List

taskListId() fallback getSessionId()
createSubagentContext() session

```
// src/utils/tasks.ts:199
export function getTaskListId(): string {
  if (process.env.CLAUDE_CODE_TASK_LIST_ID) return ...
  const teammateCtx = getTeammateContext()
  if (teammateCtx) return teammateCtx.teamName // leader task list
  return getTeamName() || leaderTeamName || getSessionId()
}
```

5.5 "8 TaskCreate"

8 V2 TaskCreateTool V1 TodoWriteTool

	V1 TodoWriteTool	V2 TaskCreateTool
		TaskUpdate
8	1 TodoWrite	8 TaskCreate

8 TaskCreate AI TaskCreate
ID

```
AI
TaskCreate({subject: "1", description: "..."})
TaskCreate({subject: "2", description: "..."})
TaskCreate({subject: "3", description: "..."})
...

TaskUpdate({taskId: "2", addBlockedBy: ["1"]})
TaskUpdate({taskId: "3", addBlockedBy: ["1"]})
```

6.

V2

V2

```

Pending
( )
In Progress
(+ )
```

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- `lock.lock()`
- `lock.unlock()`

■■■■■■■■■■ ID ■■■■■■■■■■ high water mark ■■■

```

##### #3 #####
  # # #
  ▼ ▼ ▼
##: 1.json 2.json ##: 1.json 2.json ##: 1.json 2.json 4.json
##: 0 ##: 0 ##: 3 ##: 3
↑ ↑
##### # ID = ## + 1 = 4

```

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```
■■■■■■■■■■■■■■■■■■■■bash ■■■agent ■■■■■■■■■■■■■■■■
```

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7 ■■■■■

```
// src/Task.ts:6-13
export type TaskType =
  | 'local_bash' // ■■ bash ■■
  | 'local_agent' // ■■ agent ■■■
  | 'remote_agent' // ■■ agent
  | 'in_process_teammate' // ■■■■■
  | 'local_workflow' // ■■■■■
  | 'monitor_mcp' // MCP ■■
  | 'dream' // ■■■■
```



```
pollTasks() src/utils/task/framework.ts:255
generateTaskAttachments()
```

```

##### 1: #####
##### → processUserInput.ts:504
  █→ getAttachmentMessages() → getAttachments() → #####
  getUnifiedTaskAttachments() → generateTaskAttachments()

##### 2: #####
queryLoop ##### → query.ts:1580
  █→ getAttachmentMessages() → getAttachments() → #####
  getUnifiedTaskAttachments() → generateTaskAttachments()

```

AI

```

■■■■■ ■■→ processUserInput ■■→ ■■■■■■ (■■■■■)
■
■ (AI ■■■■■)
▼
queryLoop ■■■■■
■→ ■■■■■ #1 ■■→ ■■■■■ ■■→ ■■■■■■
■→ ■■■■■ #2 ■■→ ■■■■■ ■■→ ■■■■■■
■→ ■■■■■ #3 ■■→ ■■■■■ ■■→ ■■■■■■
■→ ...

```

POLLINTERVALMS = 1000 framework.ts:22 pollTasks() generateTaskAttachments()

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```
// framework.ts:213
export function applyTaskOffsetsAndEvictions(
  setAppState: SetAppState,
  updatedTaskOffsets: Record<string, number>,
  evictedTaskIds: string[],
): void {
  setAppState(prev => {
    // ■■■■■■■■■■
    for (const id of offsetIds) {
      const fresh = newTasks[id]
      if (fresh?.status === 'running') { // ■■ still running ■■■■■■
        newTasks[id] = { ...fresh, outputOffset: updatedTaskOffsets[id]! }
      }
    }
    for (const id of evictedTaskIds) {
      const fresh = newTasks[id]
      if (!fresh || !isTerminalTaskStatus(fresh.status) || !fresh.notified) continue
    }
  })
}
```

```
delete newTasks[id]
}
})
}
```



```
// src/utls/task/diskOutput.ts
export const MAX_TASK_OUTPUT_BYTES = 5 * 1024 * 1024 * 1024 // 5GB ■■
```

- `O_NOFOLLOW` `Unix`
- `+` `drain`
- `5GB` [output truncated: exceeded 5GB disk cap]
- `(getTaskOutputDelta)`

■ generateTaskAttachments ■■■■

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8. ■■■■■ UI ■■

AppState ■■■■

```
// src/state/AppStateStore.ts:220
todos: { [agentId: string]: TodoList } // V1 todo ■■

// AppStateStore.ts:160
tasks: { [taskId: string]: TaskState } // ■■■■■■■■ V2 todo■
```

React

```
// src/state/AppState.tsx
export function useAppState<T>(selector: (state: AppState) => T): T
export function useSetAppState(): (updater: (prev: AppState) => AppState) => void
```

- useAppState —
- useSetAppState — 

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```
██ /tasks ██████
```

```
// src/commands/tasks/tasks.tsx
export async function call(onDone, context) {
  return <BackgroundTasksDialog toolUseContext={context} onDone={onDone} />
}
```

9.

V2

```
~/claude/tasks/
■■■■ <taskListId>/ ← ■■■■■■
■ ■■■■ .lock ← ■■■■
■ ■■■■ .highwatermark ← ■■ ID ■■
■ ■■■■ 1.json ← ■■ #1
■ ■■■■ 2.json ← ■■ #2
■ ■■■■ ...
■■■■ ...
```

taskListId ■■■■■■

```
1. CLAUDE_CODE_TASK_LIST_ID █████
2. █████ teamName
3. getTeamName() █████
4. leaderTeamName █████
5. Session ID █████
```

```
■■■■■■■■■■ CLAUDECODETEAM_NAME ■■■■■■■■■■getTeamName() ■■■■■■■■■■  
■■■■■■■■■■
```

```
<project_temp>/<sessionId>/tasks/  
■■■ b8n3x123.output ← bash ■■■■  
■■■ a9m2x456.output ← agent ■■■■  
■■■ ...
```

- `O_EXCL`
- `O_NOFOLLOW`
- session ID

10.

V1 TodoWriteTool

Journal Pre-proof

```
// LLM ██████
TodoWrite({
  todos: [
    { content: "██████", status: "in_progress", activeForm: "██████" },
    { content: "██████", status: "pending", activeForm: "██████" },
    { content: "██████", status: "pending", activeForm: "██████" },
  ]
})
```

V2 Task

```
// 1. ██████
TaskCreate({
  subject: "██████████",
  description: "██████████API █████JWT █████",
  activeForm: "██████████"
})
// → █ { task: { id: "1", subject: "..." } }

// 2. ██████████
TaskCreate({ subject: "███████", description: "...", activeForm: "..." })
// → █ { task: { id: "2", subject: "..." } }

TaskUpdate({ taskId: "2", addBlockedBy: ["1"] })
// → task #2 █ task #1 █
```

```
// 3. █████
TaskUpdate({ taskId: "1", status: "in_progress" })
// → █████ owner

// 4. ████████
TaskList()
// → #1 [in_progress] ██████████ (agent-name)
// → #2 [pending] ████████ [blocked by #1]

// 5. ████████
TaskGet({ taskId: "1" })
// → ████████

// 6. █████
TaskUpdate({ taskId: "1", status: "completed" })

// 7. █████
TaskUpdate({ taskId: "2", status: "deleted" })
```

11. ██████████

██	██	███
`src/tools/ToDoWriteTool/ToDoWriteTool.ts`	V1 Todo █████	███
`src/tools/ToDoWriteTool/prompt.ts`	V1 █████	███
`src/utils/todo/types.ts`	ToDoItem/ToDoList █████	██
`src/tools/TaskCreateTool/TaskCreateTool.ts`	V2 ████████	███
`src/tools/TaskCreateTool/prompt.ts`	V2 ██████████	███
`src/tools/TaskUpdateTool/TaskUpdateTool.ts`	V2 ████████	███
`src/tools/TaskUpdateTool/prompt.ts`	V2 ██████████	███
`src/tools/TaskListTool/TaskListTool.ts`	V2 █████	███
`src/tools/TaskListTool/prompt.ts`	V2 ██████████	██
`src/tools/TaskGetTool/TaskGetTool.ts`	V2 ████████	██
`src/tools/TaskGetTool/prompt.ts`	V2 ██████████	██
`src/utils/tasks.ts`	██ CRUD █ + ███	██████
`src/Task.ts`	██████████	████
`src/tasks.ts`	██████████	██
`src/utils/task/framework.ts`	████████████generateTaskAttachments/ applyTaskOffsets█pollTasks █████	██████
`src/utils/task/diskOutput.ts`	██████████	████
`src/state/AppStateStore.ts`	█████ (todos/tasks ███)	███
`src/state/AppState.tsx`	React ███ Hooks	██
`src/tools/TaskStopTool/TaskStopTool.ts`	██████████	███
`src/tools/TaskOutputTool/TaskOutputTool.tsx`	██████████	██
`src/commands/tasks/tasks.tsx`	`/tasks` ███ (███ UI)	██
`src/tasks/types.ts`	TaskState ███	██



■ ■	V1 (TodoWriteTool)	V2 (Task ■ ■ ■)	■ ■ ■ ■ ■
■ ■	■ ■ ■ ■	■ ■ ■ ■ ■ ■ ■ ■	■ ■ ■ ■ ■ ■ ■ ■
■ ■ ■ ■	LLM ■ ■ ■ ■ ■	LLM ■ ■ ■ ■ ■	■ ■ ■ ■ ■ ■ ■ ■ / ■ ■ ■ ■ ■ ■ ■ ■ ■ ■
■ ■ ■ ■	■ ■ (AppState)	■ ■ JSON ■ ■	■ ■ + ■ ■ ■ ■
■ ■ ■	■	■	■ ■ ■ ■ ■
■ ■ ■ ■	■ ■ ■	■ ■ ■	■ ■ ■ ■
■ ■ ■ ■	■ ■ ■	■ ■ ■	■ ■ ■
Feature Flag	`!isTodoV2Enabled()`	`isTodoV2Enabled()`	■ ■ ■ ■